$$\beta_u|u\rangle + \beta_d|d\rangle.$$

We assume each state is normalized:

$$\begin{aligned} \alpha_u^*\alpha_u + \alpha_d^*\alpha_d &= 1 \\ \\ \beta_u^*\beta_u + \beta_d^*\beta_d &= 1. \end{aligned} \tag{6.4}$$

And in fact these separate normalization equations for each subsystem play a crucial role in defining product states. If they did not hold, we would not have a product state. The product state describing the combined system is

$$|product\ state\rangle = \Big\{\alpha_u|u\} + \alpha_d|d\}\Big\} \otimes \Big\{\beta_u|u\rangle + \beta_d|d\rangle\Big\},$$

where the first factor represents Alice's state and the second factor represents Bob's. Expanding the product and switching to composite notation, the right-hand side becomes

$$\alpha_u\beta_u|uu\rangle + \alpha_u\beta_d|ud\rangle + \alpha_d\beta_u|du\rangle + \alpha_d\beta_d|dd\rangle. \tag{6.5}$$

The main feature of a product state is that each subsystem behaves independently of the other. If Bob does an experiment on his own subsystem, the result is exactly the same as it would be if Alice's subsystem did not exist. The same is true for Alice, of course.

Exercise 6.2: Show that if the two normalization conditions of Eqs. 6.4 are satisfied, then the state-vector of Eq. 6.5 is automatically normalized as well. In other words, show that for this product state, normalizing the overall state-vector does not put any additional constraints on the α's and β's.